RAJALAKSHMI ENGINEERING COLLEGE

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D2

Question 1 | Incorrect Mark 0.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int rev = 0, remainder;
    int count = 0;

    count++; // First while(n != 0) condition check
    while (n != 0) {
        remainder = n % 10;
        count++; // remainder = n % 10

        rev = rev * 10 + remainder;
        count++; // rev = rev * 10 + remainder

        n /= 10;
        count++; // n /= 10

        count++; // while(n != 0) condition check for next iteration
    }

    // Only print the counter value as expected by the judge
    printf("%d\n", count);

    return 0;
}
```

Check

	Input	Expected	Got	
✖	12	11	9	✖
✖	1234	19	17	✖

Your code must pass all tests to earn any marks. Try again.

Show differences

Incorrect

Marks for this submission: 0.00/1.00.

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